On the Length of a Line

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1 Why the line length matters

Most of what makes a page tiring has nothing to do with the font. It is the length of the line. Past about seventy-five characters the eye loses its place on the way back to the left margin, and readers start skipping lines without noticing.[1]

[1] Bringhurst puts the comfortable range at 45–75 characters for a single column; Butterick says much the same.

Below forty-five the opposite happens: the reader spends more time jumping between lines than reading them, and justified text develops gaps. The specimen in Figure 1 is invented, but the shape is the one every style guide describes.

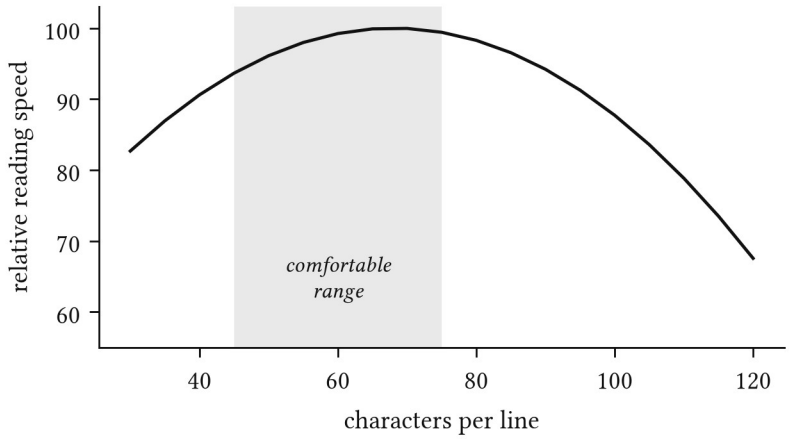


Figure 1: Reading speed against line length (illustrative curve).

1.1 What the presets do about it

Each preset fixes the text width so that 11-point type lands inside the range. The essay preset narrows it further and hands the spare width to side notes, which is where Tufte keeps references, small figures and asides that would otherwise interrupt a sentence.

Table 1: Measure by preset, measured on this specimen.

Preset	Body type	Text width	Characters / line
article	Libertine 11 pt	14.8 cm	≈ 74
journal	Libertine 10.5 pt	16.2 cm	≈ 86 (or 2 × 40)
essay	Palatino 11 pt	10.8 cm	≈ 58

2 A worked formula

A filter that combines a prior belief with one noisy reading uses nothing more than Bayes' rule:

$$P(H \mid E) = \frac{P(E \mid H)\,P(H)}{P(E \mid H)\,P(H) + P(E \mid \neg H)\,P(\neg H)} \tag{1}$$

Equation (1) is set in Computer Modern, the same face LaTeX uses, and numbered at the right margin. With a prior of 0.30 and a sensor that is right 80% of the time, one positive reading moves the posterior to *0.24/0.38 = 0.632*.

> **Info:** The posterior is lower than the sensor's 80% accuracy because the prior is low. A rare event stays fairly unlikely even after one good piece of evidence.

2.1 Code and output

Listing 1: bayes.py

```
def bayes(prior, hit, false_alarm):
    num = prior * hit
    return num / (num + (1 - prior) * false_alarm)

print(round(bayes(0.30, 0.80, 0.20), 4))
```

```
$ python bayes.py
0.6316
```

Figure 2: Program output.

3 The pipeline

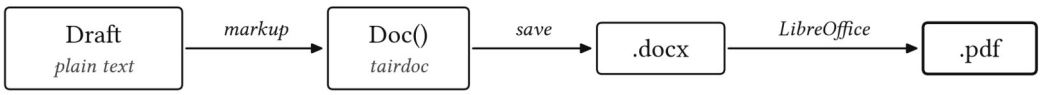


Figure 3: From draft to PDF.

1. Write the text first, in plain sentences.
2. Build it with one call per element; the preset handles spacing.
3. Open the rendered pages and fix what looks wrong before sending anything.

   Typography exists to honour content. —Robert Bringhurst